1. Regularized argmax

Core question: 15 min prep for (a)–(b), then 30 min discussion for (a)–(f).

Let $K \geq 2$, $\boldsymbol{z} \in \mathbb{R}^K$, and

$$\Delta_K = \{\boldsymbol{q} \in \mathbb{R}^K : q_i \geq 0,\ \sum_i q_i = 1\}, \qquad H(\boldsymbol{q}) = -\sum_{i=1}^{K} q_i \log q_i.$$

Use natural logarithms and $0 \log 0 = 0$.

1a) Consider

$$\max_{\boldsymbol{q} \in \Delta_K} \langle \boldsymbol{q}, \boldsymbol{z} \rangle.$$

Find the optimal value and characterize all maximizers, including when several scores tie. When is the maximizer unique?

1b) Consider the entropy-regularized problem

$$\max_{\boldsymbol{q} \in \Delta_K} \{\langle \boldsymbol{q}, \boldsymbol{z} \rangle + H(\boldsymbol{q})\}.$$

First look for an interior stationary point, so assume $q_i > 0$ and impose only $\sum_i q_i = 1$ in the Lagrangian. Derive the stationary equations, solve for q_i, normalize, and state the candidate probability vector.

1c) In (b), the constraints $q_i \geq 0$ were not put in the Lagrangian. They are still part of the problem, because the domain is Δ_K. Explain why this is safe here. Your answer should address: why the candidate from (b) cannot have a negative coordinate; what would happen if we included multipliers $\nu_i \geq 0$ for $-q_i \leq 0$; why no entropy-regularized optimum can have $q_i = 0$; and why the same shortcut will fail for the quadratic problem in (f).

1d) Give a first global proof using concavity. Start from $h(q) = -q \log q$: compute $h'(q)$ and $h''(q)$. Explain why the objective is strictly concave on the relative interior of the simplex, why a maximizer cannot have some $q_i = 0$, and why the stationary point from (b) is the unique global maximizer.

1e) Give a second global proof using KL divergence. Let $\boldsymbol{p} = \mathrm{softmax}(\boldsymbol{z})$ and $Z = \sum_j e^{z_j}$. First show that $z_i = \log p_i + \log Z$. Then prove that for any $\boldsymbol{q} \in \Delta_K$,

$$\langle \boldsymbol{q}, \boldsymbol{z} \rangle + H(\boldsymbol{q}) = \log Z - D_{\mathrm{KL}}(\boldsymbol{q} \| \boldsymbol{p}).$$

Use this to re-prove global optimality, uniqueness, and the optimal value.

1f) Take $K = 2$, $\boldsymbol{z} = (t, 0)^\top$, and replace entropy by a quadratic penalty:

$$\max_{\boldsymbol{q} \in \Delta_2} \left\{ \langle \boldsymbol{q}, \boldsymbol{z} \rangle - \tfrac{1}{2} \|\boldsymbol{q}\|_2^2 \right\}.$$

Write $\boldsymbol{q} = (p, 1-p)^\top$. Find the maximizing $p^Q(t)$, taking account of $0 \leq p \leq 1$, and identify the active constraints. Compare with $p^H(t) = \mathrm{softmax}(t, 0)_1$. Which one reaches exact 0 or 1 at finite t?

1. Extra questions

Outside lecture. Difficulty: no star = routine, * = hard, ** = very hard.

1g) For $\tau > 0$, derive

$$\boldsymbol{q}^{(\tau)}(\boldsymbol{z}) = \arg\max_{\boldsymbol{q}\in\Delta_K} \{\langle \boldsymbol{q}, \boldsymbol{z}\rangle + \tau H(\boldsymbol{q})\}.$$

Describe the limits $\tau \downarrow 0$ and $\tau \to \infty$, including tied maximum scores.

1h) For arbitrary K, define

$$\boldsymbol{q}^{Q}(\boldsymbol{z}) = \arg\max_{\boldsymbol{q}\in\Delta_K} \left\{\langle \boldsymbol{q}, \boldsymbol{z}\rangle - \tfrac{1}{2}\|\boldsymbol{q}\|_2^2\right\}.$$

Show that this is the Euclidean projection of $\boldsymbol{z}$ onto Δ_K. Use KKT to derive

$$q_i^{Q} = [z_i - \theta]_+, \qquad \sum_i [z_i - \theta]_+ = 1.$$

Check that $K = 2$, $\boldsymbol{z} = (t, 0)^\top$ recovers (f) for all t.

1i)* Let $F(\boldsymbol{z}) = \log \sum_i e^{z_i}$. Show $\nabla F(\boldsymbol{z}) = \mathrm{softmax}(\boldsymbol{z})$. Derive the Jacobian

$$\mathbf{J}(\boldsymbol{z}) = \mathrm{diag}(\boldsymbol{q}) - \boldsymbol{q}\boldsymbol{q}^\top, \qquad \boldsymbol{q} = \mathrm{softmax}(\boldsymbol{z}).$$

Using the hint that $\boldsymbol{v}^\top \mathbf{J} \boldsymbol{v}$ is a variance, prove

$$\|\,\mathrm{softmax}(\boldsymbol{z}) - \mathrm{softmax}(\boldsymbol{z}')\|_2 \le \tfrac{1}{2}\|\boldsymbol{z} - \boldsymbol{z}'\|_2.$$

Show $1/2$ is the best possible constant.

2. Training a neural network's output layer

Core question: 15 min prep for (a)–(c), then 30 min discussion for (a)–(e).

Consider

$$f_{\boldsymbol{a}}(\boldsymbol{x}) = \sum_{r=1}^{m} a_r \sigma(\boldsymbol{w}_r^\top \boldsymbol{x}), \qquad \boldsymbol{x}, \boldsymbol{w}_r \in \mathbb{R}^d.$$

The hidden weights $\boldsymbol{w}_r$ and activation σ are fixed. Only the output weights $\boldsymbol{a} \in \mathbb{R}^m$ are trained. Given $(\boldsymbol{x}_i, y_i)_{i=1}^N$, minimize

$$J(\boldsymbol{a}) = \frac{1}{2} \sum_{i=1}^{N} (f_{\boldsymbol{a}}(\boldsymbol{x}_i) - y_i)^2 + \frac{\lambda}{2} \|\boldsymbol{a}\|_2^2, \qquad \lambda > 0.$$

2a) Find a feature map $\boldsymbol{\phi} : \mathbb{R}^d \to \mathbb{R}^m$ such that $f_{\boldsymbol{a}}(\boldsymbol{x}) = \boldsymbol{a}^\top \boldsymbol{\phi}(\boldsymbol{x})$. Write its components explicitly.

2b) Differentiate $J(\boldsymbol{a})$ with respect to $\boldsymbol{a}$. Write the stationarity condition satisfied by $\boldsymbol{a}^\star$.

2c) Rearrange the stationarity condition to write

$$\boldsymbol{a}^\star = \sum_{i=1}^{N} \alpha_i \boldsymbol{\phi}(\boldsymbol{x}_i).$$

Express α_i using the prediction error at $\boldsymbol{x}_i$ and λ. Your coefficients may still involve $\boldsymbol{a}^\star$; part (e) removes this dependence.

2d) Use (c) to write

$$f_{\boldsymbol{a}^\star}(\boldsymbol{x}) = \sum_{i=1}^{N} \alpha_i k(\boldsymbol{x}_i, \boldsymbol{x}).$$

Identify $k(\boldsymbol{x}, \boldsymbol{x}')$ explicitly in terms of σ and the hidden weights. Explain why it is positive semidefinite.

2e) Let $K_{ij} = k(\boldsymbol{x}_i, \boldsymbol{x}_j)$, $\boldsymbol{\alpha} = (\alpha_i)_i$, and $\boldsymbol{y} = (y_i)_i$. Derive a linear system for $\boldsymbol{\alpha}$ involving only $\mathbf{K}$, $\boldsymbol{y}$, and λ. Compare the direct $m \times m$ weight system with the $N \times N$ kernel system. Would this kernel generally remain fixed if hidden weights were also trained?

2. Extra questions

Outside lecture. Difficulty: no star = routine, * = hard, ** = very hard.

2f) Take $d = 1$, $m = 2$, $w_1 = 1$, $w_2 = -1$, and $\sigma(u) = \max\{0, u\}$. Write $\boldsymbol{\phi}(x)$, simplify $k(x, x')$, compute $k(1, 2)$ and $k(1, -2)$, and interpret what this says about inputs with opposite signs.

2g)* Train by gradient descent:

$$\boldsymbol{a}^{(s+1)} = \boldsymbol{a}^{(s)} - \eta \nabla J(\boldsymbol{a}^{(s)}), \qquad \boldsymbol{a}^{(0)} = 0.$$

Let $\boldsymbol{p}^{(s)} = (f_{\boldsymbol{a}^{(s)}}(\boldsymbol{x}_i))_{i=1}^N$. Derive an update for $\boldsymbol{p}^{(s)}$ using only $\boldsymbol{p}^{(s)}$, $\mathbf{K}$, $\boldsymbol{y}$, λ, and η. Show that the weight iterates stay in $\operatorname{span}\{\boldsymbol{\phi}(\boldsymbol{x}_i)\}$. Assuming convergence, identify the limiting predictions.

2h)* Use the scaled random-feature model

$$f_{\boldsymbol{a}}(\boldsymbol{x}) = \frac{1}{\sqrt{m}} \sum_{r=1}^{m} a_r \sigma(\boldsymbol{w}_r^\top \boldsymbol{x}), \qquad \boldsymbol{w}_r \overset{\text{iid}}{\sim} \rho.$$

Identify the finite-m kernel. Explain informally why, as $m \to \infty$, it approaches

$$\mathbb{E}_{\boldsymbol{w} \sim \rho}[\sigma(\boldsymbol{w}^\top \boldsymbol{x}) \sigma(\boldsymbol{w}^\top \boldsymbol{x}')].$$

Why is this useful if we never explicitly construct $\boldsymbol{\phi}$?

2i)** Let $\boldsymbol{\theta}$ contain all weights and fix $\boldsymbol{\theta}_0$. Define

$$\widetilde{f}_{\boldsymbol{\delta}}(\boldsymbol{x}) = f_{\boldsymbol{\theta}_0}(\boldsymbol{x}) + \nabla_{\boldsymbol{\theta}} f_{\boldsymbol{\theta}_0}(\boldsymbol{x})^\top \boldsymbol{\delta}.$$

Fit this fixed first-order model with squared loss and $\frac{\lambda}{2}\|\boldsymbol{\delta}\|^2$. Derive its kernel and, for the two-layer network, separate output-weight and hidden-weight contributions. How must the targets be adjusted for $f_{\boldsymbol{\theta}_0}$? What happens to the hidden-weight term if all output weights are initialized at zero?

Original teaching exercises. Background: UvA ML1 public kernel slides, Martins and Astudillo (2016) on sparsemax, Jacot et al. (2018) on the neural tangent kernel.